# Maxima and Minima: Formula Sheet

## Definitions

Maxima at $a$: $f(a) > f(a \pm h)$

Minima at $a$: $f(a) < f(a \pm h)$

Critical point: $f'(c) = 0$ or $f'(c)$ does not exist

Greatest on $[a, b]$: $\max\{f(a), f(b), f(c_i)\}$

Least on $[a, b]$: $\min\{f(a), f(b), f(c_i)\}$

## Tests

First derivative: $f'$: $+ \to -$ max, $- \to +$ min

Second derivative: $f'(a) = 0$, $f''(a) < 0$ max, $f''(a) > 0$ min

$n$th derivative: $n$ even: $f^{(n)}(c) < 0$ max, $> 0$ min; $n$ odd: none

## Concavity and inflection

Concave up: $f'' > 0$, $f' \uparrow$

Concave down: $f'' < 0$, $f' \downarrow$

Concave up:

$$f\left(\frac{x_1 + x_2}{2}\right) < \frac{f(x_1) + f(x_2)}{2}$$

Inflection: $f'' = 0$ or not defined, sign change

Inflection of $f$ = extremum of $f'$

## Cubic $f = ax^3 + bx^2 + cx + d$

Derivative: $f' = 3ax^2 + 2bx + c$

Extrema exist: $b^2 - 3ac > 0$

Inflection:

$$x = -\frac{b}{3a}$$

Three distinct real roots: $f(\alpha)f(\beta) < 0$

Repeated root: $f(\alpha)f(\beta) = 0$

One real root: $f(\alpha)f(\beta) > 0$ or $b^2 - 3ac \leq 0$

$x^3 + px + q$, three distinct roots: $4p^3 + 27q^2 < 0$

$a > 0$, positive maxima: $D > 0$, $-\frac{b}{3a} > 0$, $c > 0$

$a > 0$, negative minima: $D > 0$, $-\frac{b}{3a} < 0$, $c > 0$

$a > 0$, negative maxima: $D > 0$ and ($c < 0$ or $-\frac{b}{3a} < 0$)

$a > 0$, positive minima: $D > 0$ and ($c < 0$ or $-\frac{b}{3a} > 0$)

## Mensuration

Cuboid: $V = lbh$, $S = 2(lb + bh + hl)$

Cone: $V = \frac{1}{3}\pi r^2 h$, curved $\pi rl$

Cylinder: $V = \pi r^2 h$, total $2\pi rh + 2\pi r^2$

Sphere: $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$

Sector: area $\frac{1}{2}r^2\theta$, arc $r\theta$

Prism: $V =$ base area $\times$ height

Pyramid: $V = \frac{1}{3}$ base area $\times$ height

## Standard results

Rectangle in semicircle radius $r$: $A_{\max} = r^2$

Rectangle in circle radius $a$: $A_{\max} = 2a^2$ (square)

Cylinder in sphere radius $R$:

$$h = \frac{2R}{\sqrt{3}}$$

Cone in sphere radius $R$:

$$h = \frac{4R}{3}$$

Cylinder in cone of height $h$: height $\dfrac{h}{3}$

Open cylinder, fixed volume, least surface: $h = r$

Cone, fixed slant height, max volume: $\tan\theta = \sqrt{2}$

Triangle:

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$